FILTERED HODGE COMPARISON OF L2 AND INTERSECTION COHOMOLOGY FOR SHIMURA VARIETIES

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Allowed standard prerequisites (editorial).

- Original arithmetic geometry and number theory prerequisites. Original complete bibliography and source-local definitions retained.

Introduction

The Zucker conjecture states that for a Shimura variety X its square integrable ($=L_2$) cohomology is naturally isomorphic with the intersection cohomology of its Baily-Borel compactification X^* and that this is even true if instead of constant coefficients we let these cohomology groups take their values in a locally homogeneous polarized variation of Hodge structure: $\mathbf{H}^\bullet_{(2)}(X,\mathbb{E}) \cong \mathbf{IH}^\bullet(X,\mathbb{E})$. This conjecture was settled a long time ago ([5], [11], [6]), yet one aspect remained open. Each side of the isomorphism comes with its Hodge decomposition: the representation of L_2-cohomology by harmonic forms define a Hodge decomposition of $\mathbf{H}^\bullet_{(2)}(X,\mathbb{E})$ and the theory of Morihiko Saito [9] puts one on $\mathbf{IH}^\bullet(X,\mathbb{E})$, but it was not established that under this isomorphism the two coincide. Harris and Zucker, who stated this as a conjecture in [4], 5.3, proved that the natural map $\mathbf{H}^\bullet_{(2)}(X,\mathbb{E}) \to \mathbf{H}^\bullet(X,\mathbb{E})$ is in fact a morphism of mixed Hodge structures ([4], Thm. 5.4). The purpose of this note is to show that $\mathbf{H}^\bullet_{(2)}(X,\mathbb{E}) \cong \mathbf{IH}^\bullet(X,\mathbb{E})$ is indeed an isomorphism of Hodge structures.

The proofs of the Zucker conjecture mentioned above took a local form, namely as an equality in the derived category of bounded below complexes of $\mathbb{C}$-vector spaces on X^*. The refinement which we prove here is of a similar nature, the derived category in question now being one of filtered complexes that encode in a local manner the Hodge filtration. This implies that the cohomology sheaves of successive quotients will be coherent $\mathcal{O}_{X^*}$-modules, rather than being locally constant. Our main tool are the local Hecke operators that were introduced in [5] and further developed in [6] with the express goal to prove the Zucker

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conjecture as stated above. Indeed, we will show that a minor modification can do the same job in a filtered setting. This could well have been done at the time, because the underlying technique was then already available. This also explains why much of this paper consists of revisiting and recalling work from that era.

Since we posted the first version of this paper, Mingyu Ni posted another proof of our main result [8]. This preprint also contains several other related results of interest.

1. Hodge modules

1.1. Polarizable Hodge modules. Let X be a complex manifold and $\mathbb{E}$ a polarizable variation of Hodge structures on X of weight w. We here think of $\mathbb{E}$ as local system of finite dimensional $\mathbb{Q}$-vector spaces which contains a lattice (this guarantees that the monodromy can be given by integral matrices and is quasi-unipotent when restricted to a punctured disk), but we do not consider that lattice as part of the data. Recall that the underlying holomorphic vector bundle $\mathcal{E} := \mathcal{O}_X \otimes_{\mathbb{Q}} \mathbb{E}$ then comes with a flag of holomorphic subbundles $\mathcal{F}^\bullet\mathcal{E}$ (the Hodge filtration) satisfying Griffiths transversality, i.e., the property that the flat connection $\nabla = d \otimes_{\mathbb{Q}} 1_{\mathbb{E}}$ on $\mathcal{E}$ takes $\mathcal{F}^p\mathcal{E}$ to $\Omega^1_X \otimes_{\mathcal{O}_X} \mathcal{F}^{p-1}\mathcal{E}$. This connection extends to a derivation of degree 1 of square zero in $\Omega^\bullet(\mathcal{E}) = \Omega^\bullet_X \otimes_{\mathcal{O}_M} \mathcal{E} = \Omega^\bullet_M \otimes_{\mathbb{Q}} \mathbb{E}$ and furnishes the holomorphic De Rham resolution of $\mathbb{E}_{\mathbb{C}} := \mathbb{C} \otimes_{\mathbb{Q}} \mathbb{E}$,

$$\mathbb{E}_{\mathbb{C}} \to (\Omega^\bullet_X(\mathcal{E}), \nabla).$$

Note that this resolution comes with a (Hodge) filtration $\mathcal{F}^\bullet\Omega^\bullet_X(\mathcal{E})$ by subcomplexes whose pth term is $\sum_{r+s\geq p} \Omega^r_X(\mathcal{F}^s\mathcal{E})$.

We can instead consider $(\mathcal{E}, \mathcal{F}^\bullet\mathcal{E})$ as a filtered $\mathscr{D}_X$-module, where $\mathscr{D}_X$ stands for the sheaf of holomorphic differential operators $\mathcal{O}_X \to \mathcal{O}_X$ (itself filtered by order). It thus determines an object of a derived category of filtered $\mathscr{D}_X$-modules: this is the 'Riemann-Hilbert incarnation' of the pair $(\mathcal{F}^\bullet\mathcal{E}, \nabla)$. If we ignore the filtration, then both represent $\mathbb{E}_{\mathbb{C}}$ in $D^b_c(X, \mathbb{C})$, the derived category of constructible $\mathbb{C}_X$-modules with bounded cohomology.

The Dolbeault resolution gives a fine resolution (meaning that the sheaves admit partitions of unity relative to any given open cover) of the Hodge filtered complex $(\Omega^\bullet_X(\mathcal{E}), \nabla)$ by the smooth bigraded de Rham complex

$$(\mathscr{A}^{\bullet,\bullet}(\mathbb{E}_{\mathbb{C}}), d = \partial + \bar\partial)$$

with its Hodge filtration defined in an obvious manner. So if $U \subset X$ is open, then this puts on $\mathrm{H}^k(U;\mathbb{E})$ a filtration $F^\bullet \mathrm{H}^k(U;\mathbb{E})$, with $F^p \mathrm{H}^k(U;\mathbb{E})$ the part representable by a d-closed section of $\mathcal{F}^p\mathscr{A}^{\bullet,\bullet}_X(\mathbb{E})$

over U. The meaning of this filtration is however usually obscure unless U is quite special.

This illustrates the notion of a (polarized) Hodge module in the analytic setting in its most basic form (see [9], [10]), provided we make a degree shift by putting $\mathbb{E}$ in degree $-\dim_{\mathbb{C}} X$ (this is denoted $\mathbb{E}[\dim_{\mathbb{C}} X]$).

Given a complex-analytic variety Z, then a Hodge module M on Z has two basic ingredients, the first being a particular type of element $\mathbb{M}$ of the derived category $D^b_c(Z,\mathbb{Q})$ of constructible $\mathbb{Q}_Z$-modules with bounded constructible cohomology, called a perverse sheaf. A form of the Riemann-Hilbert correspondence asserts that its complexification $\mathbb{M}_{\mathbb{C}} \in D^b_c(Z,\mathbb{C})$ is representable by a regular holonomic $\mathscr{D}_Z$-module $\mathcal{M}$. The second consists of a filtration on (the de Rham resolution of) $\mathcal{M}$ and should be regarded as a way of representing the Hodge filtration. This puts for any open subset $U \subset Z$, a Hodge filtration on $\mathrm{H}^k(U;\mathbb{M}_{\mathbb{C}})$, but here again, its meaning is unclear in general. Yet, as we will see below, there is something of interest to say if we let U run over a neighbourhood basis of a fixed point in Z. It should be clear from this sketchy description that the complexification $M_{\mathbb{C}}$ of a Hodge module is entirely given by the filtered object $\mathcal{M}$. We denote the category of Hodge modules on Z by $\mathbf{MF}(Z,\mathbb{Q})$ and its complexification (so as a quotient of $\mathbf{MF}(Z,\mathbb{Q})$) by $\mathbf{MF}(Z,\mathbb{C})$.

The notion of a polarization of a Hodge module is couched in similar terms (see [9]) and a Hodge module is called polarizable if one exists. The polarizable Hodge modules are the objects of an abelian category $\mathbf{MF}(Z,\mathbb{Q})^p$ and its complexification by $\mathbf{MF}(Z,\mathbb{C})^p$.

The polarizable Hodge modules have as their building blocks the intermediate extensions of (degree shifted) polarizable variations of Hodge structures. To explain, assume that we are given an irreducible subvariety $X^* \subset Z$ of complex dimension m and a smooth subvariety $X \subset X^*$ whose complement in X^* is a proper (closed) subvariety (which makes X open-dense in X^*) and a polarized variation of Hodge structure $\mathbb{E}$ on X as before. Writing $j : X \subset Z$ for the inclusion, then the intermediate extension $j_{!*}\mathbb{E}[m]$ of $\mathbb{E}[m]$ is defined as an object of $\mathbf{MF}(Z,\mathbb{Q})^p$. It gets its name from the fact that we have a natural factorization

$$j_!\mathbb{E}[m] \to j_{!*}\mathbb{E}[m] \to j_*\mathbb{E}[m]$$

in the category $\mathbf{MF}(Z,\mathbb{Q})^p$ and that it can be considered in the abelian category $\mathbf{MF}(Z,\mathbb{Q})^p$ as an image: it has the property that it has no $\mathbf{MF}(Z,\mathbb{Q})$-subquotients supported by a proper subvariety of X^*.

Every polarizable Hodge modules on Z is isomorphic with a direct sum of polarizable Hodge modules of this type. We therefore assume in what follows that $Z = X^*$.

The corresponding extension of $\mathbb{E}$ over X^* without the shift is the intersection complex $\mathscr{IC}^\bullet_{X^*}(\mathbb{E})$. One of Saito's theorems implies that if X^* is projective, then the intersection cohomology

$$\mathrm{IH}^k(X^*;\mathbb{E}) := \mathrm{H}^k(X^*, \mathscr{IC}^\bullet_{X^*}(\mathbb{E}))$$

has a natural polarizable Hodge structure of weight $w+k$. it also tells us that for any locally closed subvariety $i_S : S \subset X^*$, the cohomology sheaves $R^k i_S^* \mathscr{IC}^\bullet_{X^*}(\mathbb{E}))$ resp. $R^k i_S^! \mathscr{IC}^\bullet_{X^*}(\mathbb{E}))$ are variations of mixed Hodge structure over an smooth open-dense subset of S

Representation by square integrable forms. Let $\mathbb{E}$ and $j : X \subset X^*$ be as above and suppose that X is endowed with a Kähler metric. The polarization and the Hodge star operator define an anti-linear map

$$\star : \mathscr{A}_X^{p,q}(\mathbb{E}_\mathbb{C}) \to \mathscr{A}_X^{m-p,m-q}(\mathbb{E}_\mathbb{C}^\vee)$$

such that for every open $U \subset X$, the hermitian pairing

$$\langle\ ,\ \rangle : (\alpha,\beta) \in \mathscr{A}_X^{p,q}(U,\mathbb{E}_\mathbb{C}) \times \mathscr{A}_X^{p,q}(U,\mathbb{E}_\mathbb{C}) \mapsto (\sqrt{-1})^{w+m}\alpha \cup \star\beta \in \mathscr{A}_X^{m,m}(U),$$

is positive with respect to the orientation (here the expression $\alpha \cup \star\beta$ combines the cup product and the natural pairing of $\mathbb{E}_\mathbb{C}$ with its dual). This defines a notion of square integrability: for $\alpha \in \mathscr{A}_X^k(U,\mathbb{E}_\mathbb{C})$ is square integrable if $\langle\alpha,\alpha\rangle$ is integrable over U. The cohomology of the subcomplex of $\mathscr{A}^\bullet(X,\mathbb{E}_\mathbb{C})$ of forms that are together with their image under ∇ square integrable, denoted $\mathrm{H}^\bullet_{(2)}(X,\mathbb{E}_\mathbb{C})$, is what is called the square integrable cohomology of $\mathbb{E}_\mathbb{C}$. Zucker observed in [13] that if the Kähler metric on X is complete and the square integrable cohomology of $\mathbb{E}_\mathbb{C}$ is finite dimensional, then the classical Hodge theory remains valid in this (possibly) noncompact setting. In particular, $\mathrm{H}^\bullet_{(2)}(X,\mathbb{E}_\mathbb{C})$ is harmonically represented and the bigrading subsists and defines a Hodge structure on $\mathrm{H}^k_{(2)}(X,\mathbb{E}_\mathbb{C})$ of weight $k+w$. So the spectral sequence for the Hodge filtration

$$\mathrm{H}^q_{(2)}(X, \mathrm{Gr}^p_\mathcal{F}\, \Omega^\bullet_X(\mathcal{E})) \Rightarrow \mathrm{H}^\bullet_{(2)}(X,\mathbb{E}_\mathbb{C}),$$

where $\mathrm{Gr}^p_\mathcal{F}\, \Omega^\bullet_X(\mathcal{E}) = \oplus_{p'+p''=p}\Omega^{p'}_X(\mathrm{Gr}^{p''}_\mathcal{F}\, \mathcal{E})$, degenerates and endows $\mathrm{H}^\bullet_{(2)}(X,\mathbb{E}_\mathbb{C})$ with its Hodge filtration.

A presheaf complex on X^* is defined by assigning to an open subset U of X^* the subcomplex of $\mathscr{A}^\bullet(U\cap X,\mathbb{E}_\mathbb{C})$ consisting of forms that together with their image under ∇ are square integrable. Its sheafication gives

a subcomplex of $j_*\mathscr{A}_X^\bullet(\mathbb{E}_\mathbb{C})$ which we shall denote by $\mathscr{L}^\bullet_{X^*,(2)}(\mathbb{E}_\mathbb{C})$. This complex comes filtered by the Hodge filtration:

$$\mathcal{F}^p\mathscr{L}^\bullet_{X^*,(2)}(\mathbb{E}_\mathbb{C}) = \mathscr{L}^\bullet_{X^*,(2)}(\mathbb{E}_\mathbb{C}) \cap j_*\mathcal{F}^p\mathscr{A}_X^\bullet(\mathbb{E}_\mathbb{C}).$$

Our goal is to give conditions which imply that this Hodge filtered complex is an incarnation of $\mathscr{IC}^\bullet_{X^*}(\mathbb{E}_\mathbb{C}) \in \mathbf{MF}(X,\mathbb{C})^p[-m]$. The following is, in view of Zucker's harmonic representation, merely a tautology.

Observation 1.1. Suppose that X^* is complex projective, the Kähler metric is complete and that $\mathscr{L}^\bullet_{X^*,(2)}(\mathbb{E}_\mathbb{C})$ is a complex of fine $\mathcal{O}_{X^*}$-modules. If the complex endowed with its Hodge filtration represents $\mathscr{IC}^\bullet_{X^*}(\mathbb{E}_\mathbb{C})$ as an element of $\mathbf{MF}(X,\mathbb{C})^p[-\dim_\mathbb{C} X]$, then the isomorphism $\mathrm{H}^k_{(2)}(X,\mathbb{E}_\mathbb{C}) \cong \mathrm{IH}^k(X^*,\mathbb{E}_\mathbb{C})$ takes the Hodge decomposition of $\mathrm{H}^k_{(2)}(X,\mathbb{E}_\mathbb{C})$ defined by its harmonic representation to the Hodge decomposition of $\mathrm{IH}^k(X^*,\mathbb{E}_\mathbb{C})$ defined by the Hodge module interpretation.

Remark 1.2. This assumption regarding $\mathscr{L}^\bullet_{X^*,(2)}(\mathbb{E}_\mathbb{C})$ is quite strong. To see what it amounts to at a given point $s \in X^*$, choose an open neighborhood U of s in X^* which is topologically an open cone with vertex s such that $\mathrm{IH}^k(U,\mathbb{E})$ represents the local intersection cohomology $\mathrm{H}^k(i_s^*\mathscr{IC}_{X^*})$, where $i_s : \{s\} \subset X^*$. Let $\{U_n\}_{n=0}^\infty$ be a strictly decreasing open neighborhood basis of s in $U = U_0$ compatible with the cone structure, so as to ensure that the restriction maps $\mathrm{IH}^k(U;\mathbb{E}) \to \mathrm{IH}^k(U_n;\mathbb{E})$ and $\mathrm{IH}^k(U \smallsetminus \{s\};\mathbb{E}) \to \mathrm{IH}^k(U_n \smallsetminus \overline{U}_{n+1};\mathbb{E})$ are isomorphisms.

If $\mathscr{L}^\bullet_{X^*,(2)}(\mathbb{E}_\mathbb{C})$ represents $\mathscr{IC}^\bullet_{X^*}(\mathbb{E}_\mathbb{C})$, then we can then identify $\mathrm{H}^k_{(2)}(U_n,\mathbb{E}_\mathbb{C})$ with $\mathrm{IH}^k(U_n;\mathbb{E}_\mathbb{C}) = \mathrm{H}^k(i_s^*\mathscr{IC}^\bullet_{X^*}(\mathbb{E}_\mathbb{C}))$ in a way that is compatible with the restriction maps. By the theory of Hodge modules, $\mathrm{H}^k(i_s^*\mathscr{IC}^\bullet_{X^*}(\mathbb{E}))$ comes with a mixed Hodge structure and hence its complexification has a Hodge filtration. On the other hand, each $\mathrm{H}^k_{(2)}(U_n \cap X,\mathbb{E}_\mathbb{C})$ is filtered by the Hodge filtration on forms described above. If we also assume, as in the Observation 1.1 above, that the Hodge filtration on $\mathscr{L}^\bullet_{X^*,(2)}(\mathbb{E}_\mathbb{C})$ represents the one defined by the theory of Hodge modules, then the image of the Hodge filtration on $\mathrm{H}^k_{(2)}(U_n \cap X,\mathbb{E}_\mathbb{C})$ in $\mathrm{H}^k(i_s^*\mathscr{IC}^\bullet_{X^*}(\mathbb{E}_\mathbb{C}))$ under the above isomorphism should then converge to the one on $\mathrm{H}^k(i_s^*\mathscr{IC}^\bullet_{X^*}(\mathbb{E}_\mathbb{C}))$ as $n \to \infty$.

The same limiting property must hold for the compactly supported version: the Hodge filtration on $\mathrm{H}^k_{(2),c}(U_n \cap X,\mathbb{E}_\mathbb{C})$ should converge to the one on $\mathrm{H}^k(i_s^!\mathscr{IC}^\bullet_{X^*}(\mathbb{E}_\mathbb{C})$, as $n \to \infty$.

In the situations we will be dealing with, X^* has the following property.

Definition 1.1. We say that an analytic stratification of X^* which has X as its open-dense stratum is metrically locally trivial if it admits local topological trivializations which preserve strata and are such that when restricted to the opesn stratum, are quasi-isometries.

This property ensures that the cohomology sheaves of both the derived object $\mathscr{IC}^\bullet_{X^*}(\mathbb{E}_\mathbb{C})$ and the complex $\mathscr{L}^\bullet_{X^*,(2)}(\mathbb{E}_\mathbb{C})$ are locally trivial on each stratum. In that case the assumption in 1.1 regarding the Hodge filtration is equivalent for these two properties for to hold for every $s \in X^*$ and this is in fact the definition which we will use in practice. We will need a similar interpretation in the situation where we only know that these hypotheses are satisfied on the complement of a point.

Lemma 1.3. Assume that we are given $s \in X^*$, an open neighborhood U of s in X^* and a continuous proper function $d : U \smallsetminus \{s\} \to (0,\infty)$ which exhibits $(U, X \cap U)$ as an open cone with vertex s: d extends continuously to U by mapping s to ∞ and $d : (U \smallsetminus \{s\}, X \cap (U \smallsetminus \{s\})) \to (0,\infty)$ is locally trivial with the local trivializations being on $X \cap (U \smallsetminus \{s\})$ a quasi-isometry.

If the Hodge filtered complex $\mathscr{L}^\bullet_{X^*,(2)}(\mathbb{E}_\mathbb{C})$ represents $\mathscr{I}^\bullet_{X^*}(\mathbb{E}_\mathbb{C})$ on $U \smallsetminus \{s\}$, then for $0 \le n < m < \infty$ we have natural linear isomorphisms

$$\begin{aligned} \mathrm{H}^k_{(2)}(d^{-1}(n,m),\mathbb{E}_\mathbb{C}) \cong \mathrm{IH}^k(d^{-1}(n,m),\mathbb{E}_\mathbb{C}) \cong \\ \cong \mathrm{IH}^k(U \smallsetminus \{s\};\mathbb{E}_\mathbb{C}) \cong \mathrm{H}^k(i_s^* j_{s*} \mathscr{IC}^\bullet_{U\smallsetminus\{s\}}(\mathbb{E}_\mathbb{C})), \end{aligned} \tag{1.1}$$

(where $j_s : U \smallsetminus \{s\} \subset U$) and the Hodge filtration on the left defines via the above isomorphisms a family of filtrations on the right with the property that if we first let $m \to \infty$ and then let $n \to \infty$ we get the natural Hodge filtration on $\mathrm{H}^k(i_s^* j_{s*} \mathscr{IC}^\bullet_{U\smallsetminus\{s\}}(\mathbb{E}_\mathbb{C}))$ as a limit.

Proof. Recall that $\mathscr{IC}^\bullet_{U\smallsetminus\{s\}}(\mathbb{E}_\mathbb{C})$ stands for an object in a derived category of filtered complexes. We are given that the filtered complex $\mathscr{L}^\bullet_{U\smallsetminus\{s\},(2)}(\mathbb{E}_\mathbb{C})$ represents that object. Since the latter is a complex of fine sheaves, it follows that the higher direct images $R^k d_* \mathscr{L}^\bullet_{U\smallsetminus\{s\},(2)}(\mathbb{E}_\mathbb{C})$ $(k > 0)$ vanish, so that the (zeroth) sheaf direct image $d_* \mathscr{L}^\bullet_{U\smallsetminus\{s\},(2)}(\mathbb{E}_\mathbb{C})$ is also a fine complex and represents the direct image $d_* \mathscr{IC}^\bullet_{U\smallsetminus\{s\}}(\mathbb{E}_\mathbb{C}))$ in a derived category of filtered complexes on $(0,\infty)$. So the cohomology of this direct image is a graded local system on $(0,\infty)$ which we can identify with $R^\bullet d_* \mathscr{IC}^\bullet_{U\smallsetminus\{s\}}(\mathbb{E}_\mathbb{C}))$. In particular, we get the linear isomorphisms (1.1) with the first line even as isomorphisms of filtered vector spaces. It remains to observe that the definition of a Hodge module is such that the Hodge filtration on $\mathrm{IH}^k(d^{-1}(n,m),\mathbb{E}_\mathbb{C})$ has the asserted limiting property. □

2. The case of a Shimura variety

We want to apply this to the case where X is locally symmetric variety (and hence quasiprojective) and X^* the Baily-Borel compactification of X. These data involve a reductive linear $\mathbb{Q}$-group $\mathcal{G}$. We write G for the Lie group $\mathcal{G}(\mathbb{R})$ ([1]), and assume that its center $Z(G)$ is compact. So $G/Z(G)$ is an adjoint group whose symmetric space D is that of of G. We assume that D comes with a G-invariant complex structure, which makes it a hermitian domain. We further assume given a neat arithmetic subgroup $\Gamma \subset \mathcal{G}(\mathbb{Q})$. Then $X_\Gamma := \Gamma\backslash D$ is a quasi-projective complex manifold. We call such an X_Γ a locally symmetric variety. We often write X for X_Γ, when Γ is understood.

We also assume that we are given a finite dimensional $\mathbb{Q}$-representation E of $\mathcal{G}$. This gives rise to a local system $\mathbb{E} = \mathbb{E}_\Gamma$ on X (given as the quotient of the trivial local system $E\times D \to D$ by the diagonal action of Γ). It is convenient (and for what follows no loss of generality) to assume that $\mathbb{E}$ is $\mathbb{Q}$-irreducible, for this implies that $\mathbb{E}$ admits the structure of a polarizable variation of Hodge structure $\mathbb{E}$ over X of some weight w with fiber E (in our set-up E only determines the parity of w; in the Shimura setting this weight is specified by an action on E of a central extension of $\mathcal{G}$ by a $\mathbb{Q}$-split copy of $\mathbb{G}_m$). We shall refer to such an $\mathbb{E}$ as a locally homogeneous polarizable variation of Hodge structure on X. The Baily-Borel compactification $j : X \hookrightarrow X^*$ completes X to a normal projective variety. Its boundary $X^* \smallsetminus X$ comes with a stratification, all of whose members are themselves have the structure of a locally symmetric variety. More details of this construction will be recalled below.

The domain D comes with a complete G-invariant Kähler metric and this makes X a locally homogeneous Kähler manifold. Zucker showed that the natural stratification of X^* is metrically topologically trivial in the sense of Definition 1.1 and that each of the sheaves $\mathscr{L}^k_{X^*,(2)}(\mathbb{E})$ is fine ([13], Prop. (4.4)). A local form of his conjecture (proved in different ways in [5], [11], [6]) asserts that $\mathscr{L}^\bullet_{X^*,(2)}(\mathbb{E})$ is an incarnation of $\mathscr{IC}^\bullet_{X^*}(\mathbb{E}_\mathbb{C})$ in $D^b_c(X^*,\mathbb{C})$. This implies that we have natural isomorphism $\mathrm{H}^k_{(2)}(X,\mathbb{E}_\mathbb{C}) \cong \mathrm{IH}^k(X^*,\mathbb{E}_\mathbb{C})$, but as mentioned in the introduction, does not imply that the harmonic Hodge decomposition equals Saito's one. By observation 1.1 this requires that we establish a filtered local version of the Zucker conjecture in the sense that $\mathscr{L}^\bullet_{X^*,(2)}(\mathbb{E})$ with its Hodge filtration is a cohomological Hodge complex and represents $\mathscr{IC}^\bullet_{X^*}(\mathbb{E})$

[1] As a rule, we denote an algebraic group defined over $\mathbb{Q}$ by a calligraphic capital font and the Lie group defined by its group of real points by its roman counterpart.

in $\mathbf{MF}(X^*,\mathbb{C})[-\dim_{\mathbb{C}} X]$. The main result of this paper says that this is the case.

Theorem 2.1. The complex $\mathscr{L}^\bullet_{X^*,(2)}(\mathbb{E})$ is a fine $\mathcal{O}_{X^*}$-module and the Hodge filtered complex $\mathscr{L}^\bullet_{X^*,(2)}(\mathbb{E})$ represents $\mathscr{IC}^\bullet_{X^*}(\mathbb{E}_{\mathbb{C}})$ as a degree shifted complex complexified polarized Hodge module. So for every open subset $U \subset X^*$ we have an isomorphism $\mathbf{H}^k_{(2)}(U\cap X,\mathbb{E}) \cong \mathbf{IH}^k(X,\mathbb{E})$ of filtered vector spaces, which is functorial in U.

2.1. Hecke equivariance. Before we proceed to give the proof, we observe the functorial behaviour in Γ. A subgroup $\Gamma' \subset \Gamma$ of finite index determines a natural map $F : X_{\Gamma'} \to X_\Gamma$. Since Γ is neat, this map is a finite unramified covering and $\mathbb{E}_{\Gamma'}$ can be identified with $F^*\mathbb{E}_\Gamma$. In particular there is a natural map $F_*\mathbb{E}_{\Gamma'} = F_!F^*\mathbb{E}_\Gamma \to \mathbb{E}_\Gamma$. The covering F extends uniquely to a finite projective morphism $F : X^*_{\Gamma'} \to X^*_\Gamma$ and so we get natural morphisms

$$F^*\mathscr{IC}^\bullet_{X^*_\Gamma}(\mathbb{E}_\Gamma) \to \mathscr{IC}^\bullet_{X^*_{\Gamma'}}(\mathbb{E}_{\Gamma'}) \quad \text{and} \quad F_!\mathscr{IC}^\bullet_{X^*_{\Gamma'}}(\mathbb{E}_{\Gamma'}) \to \mathscr{IC}^\bullet_{X^*_\Gamma}(\mathbb{E}_\Gamma).$$

The first arrow is an isomorphism and the composite map

$$\mathscr{IC}^\bullet_{X^*_\Gamma}(\mathbb{E}_\Gamma) \to F_*F^*\mathscr{IC}^\bullet_{X^*_\Gamma}(\mathbb{E}_\Gamma) = F_!F^*\mathscr{IC}^\bullet_{X^*_\Gamma}(\mathbb{E}_\Gamma) \to \mathscr{IC}^\bullet_{X^*_\Gamma}(\mathbb{E}_\Gamma)$$

is simply multiplication by the degree of the covering. After a degree shift, this little diagram is one of polarized Hodge modules. The covering F is also a local isometry and hence $\mathscr{L}^\bullet_{X^*_\Gamma,(2)}(\mathbb{E}_\Gamma)$ and $\mathscr{L}^\bullet_{X^*_{\Gamma'},(2)}(\mathbb{E}_{\Gamma'})$ possess the same properties and induce maps between the associated spaces of harmonic forms. In particular, the validity of Theorem 2.1 for Γ' implies its validity for Γ. Indeed, that theorem can be phrased (and is probably best understood) in adelic language.

This also makes it clear that we have at our disposal an action of the Hecke algebra on both $\mathscr{IC}^\bullet_{X^*}(\mathbb{E}_{\mathbb{C}})$ and $\mathscr{L}^\bullet_{X^*,(2)}(\mathbb{E})$. Concretely, given $g \in \mathcal{G}(\mathbb{Q})$, then $g\Gamma g^{-1} \cap \Gamma$ is of finite index in Γ and $g^{-1}\Gamma g$, and hence gives rise to a finite (Hecke) correspondence

$$T_g : X_\Gamma \overset{F}{\leftarrow} X_{\Gamma\cap(g^{-1}\Gamma g)} \overset{g.}{\cong} X_{(g\Gamma g^{-1})\cap\Gamma} \overset{F'}{\longrightarrow} X_\Gamma$$

where F and F' are the natural maps and the middle isomorphism is induced by the action of g on D. This gives us endomorphisms T_g^* of the polarized Hodge module $\mathscr{IC}^\bullet_{X^*}(\mathbb{E}_{\mathbb{C}})$ (where the polarization gets multiplied by the degree of this correspondence) and the complex $\mathscr{L}^\bullet_{X^*,(2)}(\mathbb{E})$ with the latter inducing an endomorphism of the space harmonic forms. This is relevant here, because this leads us to impose natural boundary conditions (namely equivariance with respect to local Hecke operators) on the complex $\mathscr{L}^\bullet_{X^*,(2)}(\mathbb{E})$ which respect the Hodge filtration and which

are inherited by the harmonic forms. These boundary conditions were key in our original proof of the Zucker conjecture [5].

2.2. A brief review of the Baily-Borel compactification. What follows can be found in [1], but is presented here in a somewhat more geometric spirit, as for example in [3].

We recall that X^* appears as an orbit space of the Baily-Borel-Satake extension D^* of the symmetric domain D. That extension is a disjoint union of hermitian domains endowed with the so-called horocyclic topology. It contains D as an open subset and $D^* \smallsetminus D$ consists of the rational boundary components of D. The closure of each rational boundary component in D^* yields its own Baily-Borel-Satake extension. The action of G on D does not extend to D^*, but the action of $\mathcal{G}(\mathbb{Q})$ does.

The rational boundary components are in bijective correspondence with the maximal proper $\mathbb{Q}$-parabolic subgroups of $\mathcal{G}$. We first recall how this comes about.

Let $\mathcal{P} \subsetneq \mathcal{G}$ be a maximal proper $\mathbb{Q}$-parabolic subgroup. The Lie group P underlying $\mathcal{P}(\mathbb{R})$ acts transitively on D. The unipotent radical $\mathcal{R}_u(\mathcal{P})$ is nontrivial, but at most 2-step nilpotent: the center $\mathcal{U}_\mathcal{P}$ of $\mathcal{R}_u(\mathcal{P})$ is nontrivial and $\mathcal{V}_\mathcal{P} := \mathcal{R}_u(\mathcal{P})/\mathcal{U}_\mathcal{P}$ is abelian. So these are vector groups and the commutator map produces a surjective map $\wedge^2_{\mathbb{R}} \mathbf{Lie}(V_\mathcal{P}) \to \mathbf{Lie}(U_\mathcal{P})$. The reductive (Levi) quotient $\mathcal{L}_\mathcal{P} := \mathcal{P}/\mathcal{R}_u(\mathcal{P})$ acts on $\mathbf{Lie}(V_\mathcal{P})$ and $\mathbf{Lie}(U_\mathcal{P})$ and the commutator map is equivariant for these actions.

The center of $\mathcal{L}_\mathcal{P}$ contains a distinguished $\mathbb{Q}$-split copy $\mathcal{A}_\mathcal{P}$ of $\mathbb{G}_m$. This copy comes with a (character) isomorphism $\chi_\mathcal{P} : \mathcal{A}_\mathcal{P} \cong \mathbb{G}_m$ such that its action on $\mathbf{Lie}(\mathcal{U}_\mathcal{P})$ resp. $\mathbf{Lie}(\mathcal{V}_\mathcal{P})$ is with the character $\chi^2_\mathcal{P}$ resp. $\chi_\mathcal{P}$ We denote the inverse of $\chi_\mathcal{P}$ (when regarded as a morphism) by $\alpha_\mathcal{P} : \mathbb{G}_m \cong \mathcal{A}_\mathcal{P} \subset \mathcal{L}_\mathcal{P}$.

The adjoint action $\mathcal{L}_\mathcal{P}$ on $\mathcal{U}_\mathcal{P}$ leads to an almost decomposition (in the sense that intersections of factors are finite and central)

$$\mathcal{L}_\mathcal{P} = \mathcal{M}^\ell_\mathcal{P} \mathcal{A}_\mathcal{P} \mathcal{M}^h_\mathcal{P},$$

where $\mathcal{M}^h_\mathcal{P}$ acts trivially on $\mathcal{U}_\mathcal{P}$, whereas $\mathcal{M}^\ell_\mathcal{P}$ acts on $\mathcal{U}_\mathcal{P}$ with finite kernel. The Lie group $M^h_\mathcal{P} = \mathcal{M}^h_\mathcal{P}(\mathbb{R})$ has compact center and its symmetric space is in a natural manner a hermitian domain $D_\mathcal{P}$ (the complex structure is preserved by $M^h_\mathcal{P}$): this is the boundary component associated with $\mathcal{P}$ as mentioned above. The complex manifold $D_\mathcal{P}$ is also charactrized by the property that there is a natural P-equivariant holomorphic surjection

$$\pi_\mathcal{P} : D \to D_\mathcal{P}$$

whose fibers are the $M^\ell_{\mathcal{P}}A^\circ_{\mathcal{P}}$-orbits, where $A^\circ_{\mathcal{P}}$ is the identity component of $A_{\mathcal{P}} = \mathcal{A}_{\mathcal{P}}(\mathbb{R})$ (so $A^\circ_{\mathcal{P}}$ is isomorphic with the Lie group $\mathbb{R}_{>0}$, the positive reals regarded as a multiplicative group). The group $A^\circ_{\mathcal{P}}$ is also responsible for the geodesic flow in the fibers of $\pi_{\mathcal{P}}$ towards $D_{\mathcal{P}}$: for every $z \in D$, there is a unique Lie lift $\alpha_{\mathcal{P},z} : \mathbb{R}_{>0} \to P$ of $\alpha_{\mathcal{P}} : \mathbb{R}_{>0} \cong A^\circ_{\mathcal{P}}$ whose orbit through z is a geodesic which has $\pi_{\mathcal{P}}(z)$ as its limit at $+\infty$.

A central role in the definition of the horocyclic topology is played by a nonempty strictly convex self-dual cone $C_{\mathcal{P}}$ in the Lie algebra $\mathsf{Lie}(U_{\mathcal{P}})$. This cone is homogeneous: it is an $M^\ell_{\mathcal{P}}$-orbit. There is a natural P-equivariant projection $\rho_{\mathcal{P}} : D \to C_{\mathcal{P}}$. We combined this with $\pi_{\mathcal{P}}$ to get a P-equivariant map

$$(\rho_{\mathcal{P}}, \pi_{\mathcal{P}}) : D \to C_{\mathcal{P}} \times D_{\mathcal{P}}. \tag{2.1}$$

As mentioned, P acts transitively on D and it is clear that it acts on $C_{\mathcal{P}} \times D_{\mathcal{P}}$ via $L_{\mathcal{P}}$. In fact, the map (2.1) is a principal $R_u(\mathcal{U}_{\mathcal{P}})$ bundle. This also shows that the complex fiber dimension of $\pi_{\mathcal{P}} : D \to D_{\mathcal{P}}$ is $\dim \mathsf{Lie}(U_{\mathcal{P}}) + \frac{1}{2}\dim \mathsf{Lie}(V_{\mathcal{P}})$ (and that $\dim \mathsf{Lie}(V_{\mathcal{P}})$ must be even).

Let $C^+_{\mathcal{P}}$ stand for the convex hull of the rational points of the closure of $C_{\mathcal{P}}$ in $\mathsf{Lie}(U_{\mathcal{P}})$ (or equivalently, in $\mathsf{Lie}(G)$, because $\mathsf{Lie}(U_{\mathcal{P}}) \subset \mathsf{Lie}(G)$ is subspace defined over $\mathbb{Q}$). This is a convex cone whose faces are also convex cones (and open in their linear span). Any such face not equal to the vertex $\{0\}$ is of the form $C_{\mathcal{P}'}$ for some maximal proper rational parabolic group $\mathcal{P}'$. We shall write $D_{\mathcal{P}} \le D_{\mathcal{P}'}$ if $C_{\mathcal{P}'}$ is a face of $C^+_{\mathcal{P}}$ (written as $C_{\mathcal{P}'} \le C_{\mathcal{P}}$; it is therefore reasonable to think of the vertex $\{0\}$ as the face associated to $\mathcal{G}$ or to D). Let us note here that since the groups $A^\circ_{\mathcal{P}}$ and $A^\circ_{\mathcal{P}'}$ both act as scalar multiplication in $U_{\mathcal{P}'}$, they can be naturally identified.

So the union of D and the rational boundary components $\ge D_{\mathcal{P}}$ is a subset $\mathsf{Star}(D_{\mathcal{P}})$ of D^* whose constituents are indexed by the faces of $C^+_{\mathcal{P}}$. The projection $(\rho_{\mathcal{P}}, \pi_{\mathcal{P}}) : D \to D_{\mathcal{P}}$ extends to $(\rho_{\mathcal{P}}, \pi_{\mathcal{P}}) : \mathsf{Star}(D_{\mathcal{P}}) \to C^+_{\mathcal{P}} \times D_{\mathcal{P}}$.

The subgroup $\Gamma_{\mathcal{P}}$ of $\mathcal{P}$ is arithmetic (and hence a lattice in P) and acts on $D_{\mathcal{P}}$ via an arithmetic quotient group whose orbit space is a stratum S of X^*. We denote the kernel of this $\Gamma_{\mathcal{P}}$-action by $\Gamma^\ell_{\mathcal{P}} \subset \Gamma_{\mathcal{P}}$.

We recall from [1] that a core in $C_{\mathcal{P}}$ is a subset $K \subset C_{\mathcal{P}}$ which is invariant under the translations in $C_{\mathcal{P}^\ell}$ and for which there exist $p_0, p_1 \in C_{\mathcal{P}}$ such that $\Gamma^\ell_{\mathcal{P}}(p_0 + C_{\mathcal{P}}) \subset K \subset \Gamma^\ell_{\mathcal{P}}(p_0 + C_{\mathcal{P}})$. This notion does not depend on Γ and the convex hull of K is then also a core (see also [7]). In this paper, we shall use this notion in a more restricted sense: we call a core $K \subset C_{\mathcal{P}}$ a $\Gamma^\ell_{\mathcal{P}}$-core if in addition K is a $\Gamma^\ell_{\mathcal{P}}$-invariant, open,

convex and has smooth boundary. It is clear that $\lambda K = \alpha_{\mathcal{P}}(\sqrt{\lambda})K$ is then also a core for every $\lambda > 0$.

A relative version is defined in an obvious manner: a $\Gamma_{\mathcal{P}}$-core is an open $\Gamma_{\mathcal{P}}$-invariant subset $\mathcal{K}$ of $C_{\mathcal{P}} \times D_{\mathcal{P}}$ with smooth boundary which meets each fiber over $D_{\mathcal{P}}$ in a $\Gamma^{\ell}_{\mathcal{P}}$-core. So $\Omega_{\mathcal{K}} := (\rho_{\mathcal{P}}, \pi_{\mathcal{P}})^{-1}\mathcal{K}$ is then an open subset of D which is invariant under both the action of the nilpotent Lie group $R_u(\mathcal{P})$ and the geodesic flow towards $D_{\mathcal{P}}$. We extend it to a subset $\hat{\Omega}_{\mathcal{K}}$ of $\mathrm{Star}(D_{\mathcal{P}})$ by stipulating that $\hat{\Omega}_{\mathcal{K}} \cap D_{\mathcal{P}'} = \pi_{\mathcal{P}'}(\Omega_{\mathcal{K}})$ when $D'_{\mathcal{P}} \geq D_{\mathcal{P}}$. Then $\hat{\Omega}_{\mathcal{K}}$ is also invariant under $R_u(\mathcal{P})$ and the geodesic flow towards $D_{\mathcal{P}}$.

It is a fundamental fact (and in a sense the reason for introducing cores in the first place) that there exists a $\Gamma_{\mathcal{P}}$-core $\mathcal{K}$ such that every $\Gamma_{\mathcal{P}}$-orbit in $\Omega_{\mathcal{P}}$ is the intersection of $\hat{\Omega}_{\mathcal{P}}$ with a Γ-orbit. In other words, the orbit set $\hat{U}_S = \Gamma_{\mathcal{P}}\backslash\Omega$ can then be regarded as a subset of X^*. This leads us to the horocyclic topology on D^*: it is the coarsest topology which for every $\mathcal{P}$ makes the subsets $\{\hat{\Omega}_{\mathcal{K}}\}_{\mathcal{K}}$ a basis of neighborhoods of $D_{\mathcal{P}}$ in D^* and the projection $\pi_{\mathcal{P}} : \mathrm{Star}(D_{\mathcal{P}}) \to D_{\mathcal{P}}$ continuous. The justification is that if we endow X^* with the quotient topology, it becomes a locally compact Hausdorff space and that the sheaf on X^* of continuous $\mathbb{C}$-valued functions which are analytic on each stratum endows X^* with the structure of a normal analytic variety

Remark 2.2. The space $C^+_{\mathcal{P}}$ also admits a horocyclic topology (this uses the self-dual properties of its faces, see [7]); it is $\mathcal{P}^{\ell}(\mathbb{Q})$-equivariant and makes $\Gamma^{\ell}_{\mathcal{P}}\backslash C^+_{\mathcal{P}}$ a stratified locally compact Hausdorff space. There is a natural extension of $\rho_{\mathcal{P}}$ to a continuous map $\mathrm{Star}(D_{\mathcal{P}}) \to C^+_{\mathcal{P}}$ inducing a map $\Gamma_{\mathcal{P}}\backslash \mathrm{Star}(D_{\mathcal{P}}) \to \Gamma^{\ell}_{\mathcal{P}}\backslash C^+_{\mathcal{P}}$ over S of stratified spaces. We will however not need this in what follows.

Definition 2.1. We say that a neighborhood of $D_{\mathcal{P}}$ in $\mathrm{Star}(D_{\mathcal{P}})$ is Γ-regular if is of the form $\hat{\Omega}_{\mathcal{K}}$ with $\mathcal{K}$ a $\Gamma_{\mathcal{P}}$-core and is such that for some $\varepsilon \in (0,1)$ a nonempty intersection of $\hat{\Omega}_{(1-\varepsilon)\mathcal{K}}$ with a Γ-orbit is in fact a $\Gamma_{\mathcal{P}}$-orbit.

We call a neighbourhood of S in X^* which appears as the image of a Γ-regular neighbourhood of $D_{\mathcal{P}}$ in $\mathrm{Star}(D_{\mathcal{P}})$ a fundamental regular neighbourhood of S in X^*.

A fundamental regular neighbourhood $\hat{U}_S = \Gamma^{\ell}_{\mathcal{P}}\backslash\hat{\Omega}_{\mathcal{K}}$ of S in X^* has several nice properties. First of all, it is invariant under a geodesic flow towards S. To be precise, it takes $\hat{U}_S$ to $\mu\hat{U}_S := \Gamma_{\mathcal{P}}\backslash\Omega_{\mu'\mathcal{K}}$ for every $\mu \geq 1$. This defines a homeomorphism $(0,\infty) \times \partial\hat{U}_S \cong \hat{U}_S \smallsetminus S$ which extends to a stratified homeomorphism of $\hat{U}_S$ onto the open mapping cone of $\pi_S|\partial\hat{U}_S$ (which is here defined as the quotient of $(0,\infty] \times \partial\hat{U}_S$ obtained

by collapsing $\{\infty\} \times \partial\hat{U}_S \cong \partial\hat{U}_S$ along π_S). Secondly, the projection $\pi_S : \partial\hat{U}_S \to S$ is topologically locally trivial in a strata preserving manner. And thirdly, $\pi_S : U_S := \Gamma_{\mathcal{P}}\backslash\Omega_{\mathcal{K}} \to S$ factors over a fibration into compact nilmanifolds

$$U_S \xrightarrow{\rho_S} \Gamma_{\mathcal{L}_{\mathcal{P}}}\backslash\mathcal{K} \xrightarrow{\pi'_S} S, \tag{2.2}$$

where $\Gamma_{\mathcal{L}_{\mathcal{P}}}$ is the image of $\Gamma_{\mathcal{P}}$ in $L_{\mathcal{P}}$. The fiber of ρ_S through any point of U_S is naturally identified with the nilmanifold $R_u(\mathcal{P})/\Gamma_{R_u(\mathcal{P})}$ (this is a torus bundle over a torus; in particular, ρ_S is proper). The fiber of π'_S over $s \in S$ is the $\Gamma^{\ell}_{\mathcal{P}}$-orbit space of the core $\mathcal{K}_s$. There exist metrically locally trivial trivializations of $\hat{U}_S \to S$ in the sense of Definition 1.1 compatible with the geodesic flow towards S.

2.3. Proof of the Theorem 2.1. We begin with reviewing the proof of the Zucker conjecture as given in [5] and [6]. Every $g \in \mathcal{P}^{\ell}(\mathbb{Q})$ induces Hecke correspondence $T^{\mathcal{P}}_g$ of the germ of X^* at S, which is proportional to the identity over S. It is a local isometry which lifts to $\mathbb{E}$. So if $i_S : S \subset X^*$ is the inclusion, then $T^{\mathcal{P}}_g$ induces an endomorphism $T^{\mathcal{P}*}_g$ of the mixed Hodge module $i^*_S j_{!*}\mathbb{E}$, the intersection complex $i^*_S \mathscr{IC}^{\bullet}_{\Omega}(\mathbb{E})$ and $i^*_S \mathscr{L}^{\bullet}_{\Omega\smallsetminus S,(2)}(\mathbb{E})$.

In case g is such that $g\Gamma_{\mathcal{P}}g^{-1} \subset \Gamma_{\mathcal{P}}$, then it is the identity over S and we can choose the Γ-core $\mathcal{K}$ such that g maps a $\mathcal{K}$ to itself and hence defines in fact a finite map from a fundamental regular neighborhood $\hat{U}_S$ as above onto neighborhood of S contained in $\hat{U}_S$.

For example, if we take $g \in \mathcal{R}_u(\mathcal{P})(\mathbb{Q})$, then $T^{\mathcal{P}}_g$ preserves each fiber of ρ_S. Insofar the action on differential forms is concerned, we may take these forms in such a manner that their pull-back to D is invariant under the nilpotent Lie group $R_u(\mathcal{P})$ (the harmonic forms will automatically have this property). The Hecke action of $T^{\mathcal{P}*}_g$ on such forms is trivial. This explains why for most of its uses, $T^{\mathcal{P}*}_g$ only depends on the image $\bar{g}$ of g in the Levi quotient.

Of special interest here is the case when $\bar{g} \in \mathcal{A}_{\mathcal{P}}(\mathbb{Q})$. If $\chi_{\mathcal{P}}(\bar{g}) = \sqrt{q}$ (or equivalently, $\alpha_{\mathcal{P}}(\sqrt{q}) = \bar{g}$), then $g\Gamma_{\mathcal{P}}g^{-1} \subset \Gamma_{\mathcal{P}}$, provided q is a positive square such that $q-1$ is sufficiently divisible. We assume this is the case and then write $T^S(q)$ for $T^{\mathcal{P}}_g$. The action of $T^S(q)$ on U_S is geometrically easy to understand in terms of the factorization (2.2): since g acts as multiplication by q resp. $\sqrt{q}$ on $\mathrm{Lie}(U_{\mathcal{P}})$ resp. $\mathrm{Lie}(V_{\mathcal{P}})$, the map $T^S(q)$ takes $[\Gamma^{\ell}_{\mathcal{P}}y] \in \Gamma^{\ell}_{\mathcal{P}}\backslash\mathcal{K}$ to $[\Gamma^{\ell}_{\mathcal{P}}qy] \in \Gamma^{\ell}_{\mathcal{P}}\backslash\mathcal{K}$ with the map on their nilmanifold fibers being an isogeny: recall that the nilmanifold is a torus bundle and that over a torus and $T^S(q)$ acts in the base torus as multiplication by $\sqrt{q}$ and in the fiber torus as multiplication by q. So $T^S(q)$ induces

on the germ of X^* at S a morphism of degree

$$q^{\dim \mathrm{Lie}(U_{\mathcal{P}})}(\sqrt{q})^{\dim \mathrm{Lie}(V_{\mathcal{P}})} = q^{\dim \mathrm{Lie}(U_{\mathcal{P}})+\frac{1}{2}\dim \mathrm{Lie}(V_{\mathcal{P}})} = q^d,$$

where d is the complex codimension of S in X^*. The geodesic flow away from S produces an isotopy of $T^S(q)$ with a self map of $\hat{U}_S$ which is the identity over $\Gamma_{\mathcal{L}_{\mathcal{P}}}\backslash\mathcal{K}$. This self map then induces in each in each fiber over $\Gamma_{\mathcal{L}_{\mathcal{P}}}\backslash\mathcal{K}$ an isogeny as above. Both the singular cochain complex and the $\mathbb{E}$-valued de Rham complex of the fiber through a given point are naturally quasi-isomorphic with the Lie algebra subcomplex

$$\mathrm{Hom}_{\mathbb{R}}(\wedge^\bullet \mathrm{Lie}(R_u(\mathcal{P}), E). \tag{2.3}$$

The action of $T^S(q)$ is in this subcomplex given by the lift $g \in \mathcal{P}(\mathbb{Q})$ of $\bar{g} = \alpha_{\mathcal{P}}(\sqrt{q})$ and hence is semisimple with eigenvalues powers of $\sqrt{q}$. This is of course merely the evaluation of the action of a one-parameter subgroup on this Lie algebra complex. In this form it already appeared earlier in the work of Borel [2] and Zucker [14]. It shows that the total direct image of $\mathbb{E}|U_S$ on $\Gamma_{\mathcal{L}_{\mathcal{P}}}\backslash\mathcal{K}$ decomposes in the derived category according to the integral powers of $\sqrt{q}$. This implies the semisimplicity of $T^S(q)$-actions on various cohomology groups.

Let $s \in S$, denote by $U_s \subset \hat{U}_s$ the fiber of $U_S \subset \hat{U}_S$ over s and put $\mathbb{E}_s := \mathbb{E}|U_s$. By the defining properties of the intersection complex,

$$\mathrm{IH}^k(\hat{U}_s, \mathbb{E}_s) = \begin{cases} \mathrm{IH}^k(\hat{U}_s \smallsetminus \{s\}, \mathbb{E}_s) & \text{if } k < d, \\ 0 & \text{if } k \geq d. \end{cases}$$

$$\mathrm{IH}^k_c(\hat{U}_s, \mathbb{E}_s) = \begin{cases} \mathrm{IH}^k_c(\hat{U}_s \smallsetminus \{s\}, \mathbb{E}_s) & \text{if } k > d, \\ 0 & \text{if } k \leq d \end{cases}$$

and the polarization pairing on $\mathbb{E}_s$ induces the Poincaré duality pairing

$$\mathrm{IH}^k(\hat{U}_s \smallsetminus \{s\}, \mathbb{E}_s) \times \mathrm{IH}^{2d-k}_c(\hat{U}_s \smallsetminus \{s\}, \mathbb{E}_s) \to \mathrm{IH}^{2d}_c(\hat{U}_s \smallsetminus \{s\}; \mathbb{Q})(\cong \mathbb{Q}).$$

The pair $(\hat{U}_s, \mathbb{E}_s)$ comes with an endomorphism $T^s(q)$ of degree q^d which acts semisimply on $\mathrm{IH}^\bullet(\hat{U}_s \smallsetminus \{s\}, \mathbb{E}_s)$ with eigenvalues powers of $\sqrt{q}$. The pairing is $T^s(q)$-equivariant, where $T^s(q)$ acts on $\mathrm{IH}^{2d}(\hat{U}_s \smallsetminus \{s\}; \mathbb{Q})$ as multiplication by q^d. We restate this in terms of (a derived category of) sheaf complexes: letting j_S stand for the inclusion $\hat{U}_S \smallsetminus S \subset \hat{U}_S$, then

$$i_S^* j_{S*} \mathscr{IC}^\bullet_{\hat{U}_S \smallsetminus S}(\mathbb{E}) = i_S^* \mathscr{IC}^\bullet_{\hat{U}_S}(\mathbb{E}) \oplus i_S^! \mathscr{IC}^\bullet_{\hat{U}_S}(\mathbb{E})[-1], \tag{2.4}$$

with the first summand living in degrees $< d$ and the second in degrees $> d$. This comes a perfect duality pairing (in a derived category)

$$i_S^* j_{S*} \mathscr{IC}^\bullet_{\hat{U}_S \smallsetminus S}(\mathbb{E}) \otimes_{\mathbb{Q}_S} i_S^! j_{S*} \mathscr{IC}^\bullet_{\hat{U}_S \smallsetminus S}(\mathbb{E}) \to \mathbb{Q}_S(-d)[-2d].$$

The following is proved in [5] (Prop. 3.8) (see also [6], Thm. 3.2). The semisimplicity of the action of $T^S(q)$ on $R^\bullet i_S^* j_{S*}\mathscr{IC}^\bullet_{\hat{U}_S\smallsetminus S}(\mathbb{E})$ is not stated there explicitly, but follows from the preceding discussion.

Proposition 2.3. The action of the local Hecke operator $T^S(q)$ on the local system $R^\bullet i_S^* j_{S*}\mathscr{IC}^\bullet_{\hat{U}_S\smallsetminus S}(\mathbb{E})$ is semisimple with eigenvalues that are integral powers of q and the summand associated with the eigenvalue q^l is a locally homogeneous polarized Hodge structure on S of weight $w+\ell$.

Furthermore, $R^\bullet i_S^*\mathscr{IC}^\bullet_{\hat{U}_S}(\mathbb{E})$ resp. $R^\bullet i_S^!\mathscr{IC}^\bullet_{\hat{U}_S}(\mathbb{E})[-1]$ is the subsum of the summands with eigenvalue $< q^d$ resp. $> q^d$ (so the eigenvalue q^d does not occur). □

The proof of our main theorem will also use the following simple linear algebra observation, stated here as a lemma.

Lemma 2.4. Let T be semisimple transformation of a finite dimensional $\mathbb{C}$-vector space V whose eigenvalues are integral powers of q. Then for every subspace $F\subset V$ the limit $F_\infty := \lim_{n\to\infty} T^n F$ (taken in the Grassmannian of V) is of the form $\oplus_\ell F_{\infty,\ell}$ with $F_{\infty,\ell}$ is contained in the q^ℓ-eigenspace V_ℓ of T and obtained as the image of $F\cap\oplus_{\ell'\le\ell}V'_\ell$ under the projection onto V_ℓ. □

Proof of Theorem 2.1. Let us make the inductive assumption that it has been established that the Hodge filtered complex $\mathscr{L}^\bullet_{X^*(2)}(\mathbb{E})$ represents $\mathscr{IC}^\bullet(\mathbb{E})$ on the union of strata of complex codimension $< d$. The induction step consists in proving that this is then also true along a codimension d stratum S as above.

Our induction assumption implies that the complex $i_S^* j_{S*}\mathscr{L}^\bullet_{\hat{U}_S\smallsetminus S,(2)}(\mathbb{E})$ represents $i_S^* j_{S*}\mathscr{IC}^\bullet_{\hat{U}_S\smallsetminus S}(\mathbb{E})$. As mentioned, Zucker proved that $\pi_S : \hat{U}_S\to S$ admits metric local trivializations in the sense of Definition 1.1 that are compatible with the geodesic flow towards S (the Theorem in §3 of [14]). So $\mathscr{L}^\bullet_{X^*,(2)}(\mathbb{E})$ is locally constant along S and the cohomology of the stalk of $i_S^* j_{S*}\mathscr{L}^\bullet_{\hat{U}_S\smallsetminus S,(2)}(\mathbb{E})$ at $s\in S$ is also that of its form restriction to the fiber over s, i.e., the stalk of $R^k j_{s*}\mathscr{L}^\bullet_{\hat{U}_s\smallsetminus\{s\},(2)}(\mathbb{E}_s)$ at $s\in S$ (where $j_s : \hat{U}_s\smallsetminus\{s\}\subset\hat{U}_s$). We are then in a situation to which Lemma 1.3 applies: it tells us that for $1\le\lambda<\mu$ we have natural isomorphism of vector spaces

$$\mathrm{H}^\bullet_{(2)}(\lambda\hat{U}_s\smallsetminus\mu\hat{U}_s,\mathbb{E}_s)\cong \mathrm{IH}^\bullet(\lambda\hat{U}_s\smallsetminus\mu\hat{U}_s,\mathbb{E}_s)\cong$$
$$\cong \mathrm{IH}^\bullet(\hat{U}_s\smallsetminus\{s\},\mathbb{E}_s)\cong \mathrm{H}^k(i_s^* j_{s*}\mathscr{IC}^\bullet_{\hat{U}_s\smallsetminus\{s\}}(\mathbb{E}))$$

with the first row an isomorphism of filtered vector spaces. We get the Hodge filtration on the last term from (the image of) the Hodge filtration of the first term by first taking the limit for $\mu \to \infty$ and then the limit for $\lambda \to \infty$. The endomorphism $T^s(q)$ acts on the last term (and preserves its Hodge filtration). If we transfer this action to the top row, we can make it geometrically explicit by noting that for $1 \le \lambda < \mu$, the restriction map

$$\mathrm{H}^k_{(2)}(\lambda\Omega_s \setminus \mu\Omega_s, \mathbb{E}_s) \to \mathrm{H}^k_{(2)}(\lambda q\Omega_{s,r} \setminus \mu q\Omega_s, \mathbb{E}_s) \tag{2.5}$$

is an isomorphism and that $T^s(q)$ induces a map in the opposite direction, so that we get an endomorphism of $H^\bullet_{(2)}(\lambda\Omega_{s,n} \setminus \mu\Omega_s, \mathbb{E}_s)$. This is indeed the one coming from $\mathrm{H}^k(i_s^* j_{s*}\mathscr{IC}^\bullet_{\hat{U}_s\setminus\{s\}}(\mathbb{E}))$. It follows that we get the limit of the Hodge filtration on the stalk of $R^k j_{s*}\mathscr{L}^\bullet_{\hat{U}_s\setminus\{s\},(2)}(\mathbb{E}_s)$ at s, by taking the limit of the Hodge filtration on $\mathrm{H}^\bullet_{(2)}(\lambda U_s \setminus \mu U_s, \mathbb{E}_s)$ under the positive powers of $T^s(q)$.

Lemma 2.4 applied to this situation tells us that if we regard $H^\bullet_{(2)}(\lambda U_s \setminus \mu U_s, \mathbb{E}_s)_\ell$ as a subquotient of $H^\bullet_{(2)}(\lambda U_s \setminus \mu U_s, \mathbb{E}_s)$ (rather than as a direct summand), then its Hodge filtration reproduces the Hodge filtration of $\mathrm{IH}^\bullet(\hat{U}_s \setminus \{s\}, \mathbb{E}_s)$. In particular, it defines on $H^\bullet_{(2)}(\lambda U_s \setminus \mu U_s, \mathbb{E}_s)_\ell$ a pure Hodge structure of weight $m + \ell$. Since the eigenvalue q^d does not occur, Zucker's weight computation (formula (8) of the Theorem in §3 of [14]) allows us to conclude that the restriction map

$$\oplus_{\ell<d} H^\bullet_{(2)}(\lambda U_s \setminus q\lambda U_s, \mathbb{E}_s)_\ell \to H^\bullet_{(2)}(\lambda U_s, \mathbb{E}_s)$$

is an isomorphism. Hence

$$H^\bullet_{(2)}(\lambda U_s, \mathbb{E}_s) = \oplus_{\ell<d} H^\bullet_{(2)}(\lambda U_s, \mathbb{E}_s) \cong \oplus_{\ell<d} \mathrm{IH}^\bullet(\hat{U}_s, \mathbb{E}_s) = \mathrm{IH}^\bullet(\hat{U}_s, \mathbb{E}_s)$$

becomes is an isomorphism of Hodge structures if we let $\lambda \to \infty$. In the same way we find the corresponding property for the compactly supported version:

$$H^\bullet_{(2),c}(\lambda U_s, \mathbb{E}_s) = \oplus_{\ell>d} H^\bullet_{(2),c}(\lambda U_s, \mathbb{E}_s) \cong \oplus_{\ell>d} \mathrm{IH}^\bullet_c(\hat{U}_s, \mathbb{E}_s) = \mathrm{IH}^\bullet_c(\hat{U}_s, \mathbb{E}_s)$$

This completes the induction step and thereby finishes the proof of Theorem 2.1. □

References

[1] A. Ash, D. Mumford, M. Rapoport, Y.-S. Tai: Smooth compactifications of locally symmetric varieties (2nd ed. with the coll. of Peter Scholze). Cambridge U.P., Cambridge, 2010, x+230 pp. 9, 10

[2] A. Borel, N. Wallach: Continuous cohomology, discrete subgroups, and representations of reductive groups, Annals of Mathematics Studies 94, Princeton 1980. 13

[3] J. Chen, E. Looijenga: The homotopy type of the Baily-Borel and allied compactifications, Homology Homotopy Appl. 23 (2021), 95–119. 9
[4] M. Harris, S. Zucker: Boundary cohomology of Shimura varieties. III. Coherent cohomology on higher-rank boundary strata and applications to Hodge theory, Mém. Soc. Math. Fr. (N.S.) 85 (2001), vi+116 pp. 1
[5] E. Looijenga: L^2-cohomology of locally symmetric varieties, Compositio Math. 67 (1988), 3–20. 1, 7, 9, 12, 14
[6] E. Looijenga, M. Rapoport: Weights in the local cohomology of a Baily-Borel compactification, in Complex geometry and Lie theory. 223–260, Proc. Sympos. Pure Math. 53, Amer. Math. Soc., Providence, RI, 1991. 1, 7, 12, 14
[7] E. Looijenga: Discrete automorphism groups of convex cones of finite type, Compos. Math. 150 (2014), 1939–1962. 10, 11
[8] Ni, Mingyu: Weighted Cohomology, Hodge Theory and Intersection Cohomology of Shimura varieties, https://arxiv.org/pdf/2603.24464 2
[9] M. Saito: Modules de Hodge polarisables, Publ. Res. Inst. Math. Sci. 24 (1988), 849–995. 1, 3
[10] M. Saito: A young person's guide to mixed Hodge modules, in Hodge theory and L_2-analysis, 517–553, Adv. Lect. Math. 39, Int. Press, Somerville, MA, 2017. 3
[11] L. Saper, M. Stern: L_2-cohomology of arithmetic varieties, Ann. of Math. 132 (1990), 1–69. 1, 7
[12] S. Zucker: Hodge theory with degenerating coefficients: L_2 cohomology in the Poincaré metric. Ann. of Math. 109 (1979), 415–476.
[13] S. Zucker: L_2 cohomology of warped products and arithmetic groups, Invent. Math. 70 (1982/83), 169–218. 4, 7
[14] S. Zucker: L_2-cohomology and intersection homology of locally symmetric varieties, II, Comp. Math., 59(1986), 339–398. 13, 14, 15